

[Collaborative Docs](#) >

Discussion Agenda 02/05/2016 (High-resolution DM-only Run)

Below is a discussion agenda / live progress report for the 8th Web Conference of the AGORA Project (Feb. 5, 9 am PST/6 pm CET). This telecon will particularly focus on the progress and future work for AGORA Paper 2. Some items below were discussed amongst the Paper Group leaders on Jan. 21 in preparation of this Group-wide meeting. This Feb. 5 meeting was then followed by another smaller meeting on Feb. 18 only with code leaders joining.

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Check out the important items in brown color below!

(1st revision: 02/05/2016, 2nd revision: 03/01/2016, 3rd revision: 03/05/2016)

[1. Progress Report \(as of Feb. 2016\)](#)

We are actively making progress, chiefly, in three different areas below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																																																						
GOALS	Run Proof-of-concept-DM (~1e11 Ms halo) at higher resolution	- Implement and test common astrophysics package - Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	Calibrate SF/Fbck recipes using common disk ICs																																																						
LEADERS	- Oliver Hahn (OCA, Nice) - Christine Simpson (HITS)	- Nick Gnedin (Fermilab) - Piero Madau (UCSC) - Britton Smith (Edinburgh)	- Oscar Agertz (Surrey) - Romain Teyssier (Zurich) - James Wadsley (McMaster)																																																						
PARTICIPANTS	- Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) - Code representatives:	- Working Groups engaged: - I (common physics) - III (common cosmo ICs) - Code representatives:	- Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) - Code representatives:																																																						
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PROGRESS MADE OR BEING MADE	• Please visit Paper 2's Workspace where Oliver Hahn et al. have been updating the progress. • See Section 2 below for more items to discuss today and the 2016 timeline.	• Either via Britton's original interface or through Nick's API, GRACKLE is now fully implemented (equilibrium cooling version) in all 9 participating codes. • The resulting cooling functions is compared in isolated disk runs, Paper "4" (either in cooling only + noSF mode, or in cooling + SF + ThermalFbck mode).	• Please visit Paper 4's Workspace where Oscar Agertz et al. have been updating the progress. • All 9 participating codes have updated their "first" datasets (i.e. Grackle + noSF runs) since the last Workshop. The latest composite images on their Workspace were created on Dec. 14, 2015. • 7 code groups have submitted their "second" datasets as of Jan. 2016. Groups are still working to make Grackle work nicely with star formation and feedback. Please visit their Workspace for up-to-date information. • The agora-analysis-script repository is being updated as the analysis progresses.																
OTHER LINKS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. • MUSIC Google group	• GRACKLE Google group • GRACKLE download & documentation • AGORA API for GRACKLE by Nick Gnedin	• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. • yt-AGORA Google group																

2. Paper Group 2 - Plan For The Next Months

- Below are abbreviated talking points only for this telecon. The writings below will not be updated as the ideas in Paper "2" evolve in time.

- Participants of Paper "2" are **STRONGLY** recommended to check out and follow the up-to-date consensus found in the [Paper 2 collaborative doc](#).

(1) Feb. 5: FIRST WEB CONFERENCE

(1-1) Discussion Item #1 – Logistics

- Given the time constraint, can we narrow or redefine the scope of Paper to quickly maneuver ourselves towards the goal in the next couple of months (i.e. form a close-knit <5 code team, more compact than the 9 code cohort in the Flagship Paper)?
[Agreed. Check the table above to see the participating codes (*to be updated*).]
- For the same reason, may we opt to focus on <3 key themes that are the most scientifically rewarding and exciting, rather than trying to cover the entire analysis shopping list (i.e. identify runtime configurations such as force resolution that makes codes compatible)?
[Agreed. Detailed science objectives including numerical resolution to be used, will be decided among code leaders as we move forward.]
- Identify one *devoted* person per each code who can dedicate manpower and computing resources in the next couple of months to run high-resolution DM-only simulations, and compare the results. Confirm the leadership.
[Check the table above to see the code leaders who have committed themselves (*to be updated*). If you'd like to lead an yet uncommitted code, please let Ji-hoon know.]
- Ratify an achievable, near-term timeline.
[Near-term schedule agreed. Biweekly telecons will be held amongst code leaders.]

(1-2) Discussion Item #2 – Requirements for Initial Datasets

- See [the Paper 2 collaborative doc](#) for already discussed details.
- Any revisions (target halo mass, mask, target redshift, storage policy, etc.)?
[No revision needed for now. Target halo reconfirmed = $\sim 10^{11} M_{\text{sun}}$ halo with a quiescent merger history. Its ellipsoidal IC has been on the [webpage](#) already for a while. Rockstar halo finder will be employed.]

(2) Feb. 5 – Feb. 29: INITIAL DATASET SUBMISSION

- Release all of the 8 ellipsoidal ICs on the AGORA webpage, accompanied by a short note on astro-ph.
[Agreed. To be completed by Hahn, Onorbe, and Simpson.]
- Run a few initial tests with an AMR code to fully understand simulations' behavior with different refinement criteria at lower(?) resolution.
[Agreed. To be completed by Hahn and Simpson.]
- Then using the parameter sets identified above, run and submit first datasets for comparison and let Ji-hoon know. Biweekly telecons amongst code leaders to force ourselves.
[Agreed. Initially submit only one snapshot at $z=0$. But make sure to store as many snapshots as possible for future analysis, following the guideline in [here](#) or [here](#). If you don't have a user account at NERSC, visit [here](#).]
- Ji-hoon is willing to volunteer to collect the datasets for comparison (eventually at NERSC), and run some initial cross-platform analysis. The [agora-analysis-script](#) repository will be updated as the analysis progresses. More volunteers will be certainly welcome!

(3) Mar. 1 – Apr. 15: ADDITIONAL ITERATION

- Produce multi-panel images and plots on the web, and iterate until each code group arrives at a parameter configuration that makes its code compatible with others. Perform parameter studies as needed.

(4) Mid-to-Late April: SECOND GROUP-WIDE WEB CONFERENCE

- More to come: paper writing, etc.

[3. Other Discussion Items](#)

(1) AGORA Allocation at NERSC, PRAC Application at NCSA (?):

The AGORA storage at NERSC is renewed and guaranteed for 2016. We may decide to apply for some simulation computing time at Blue Waters@NCSA. To be further discussed amongst the Steering Committee members.

(2) Progress on Paper 4:

Please visit [this link](#) for most recent updates.

[4. Recaps](#)

(1) AGORA SMBH Prescription (Accretion and Feedback):

Please visit [this link](#).

(2) Strategy to Store Simulation Outputs:

Please visit [this link](#).

(3) yt-Sunrise Pipeline for AGORA on NERSC:

Please visit [this link](#).

(4) Mini-workshop Opportunities:

Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(5) Useful Links:

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!